

FLCR-02 - The LCR Carrier

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper defines the LCR carrier as the canonical radius-1 local readout with a distinguished center and two ordered boundary slots. The carrier is finite, inspectable, and minimal: any smaller carrier collapses either the center or one boundary address. The paper proves the binary eight-state surface, reversal action, shell grading, and shell-2 doublet-plus-pivot structure used by later papers.

Keywords

LCR carrier; radius-1 cellular automaton; C2 action; shell grading; minimal arity.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **The LCR Carrier**. The paper's immediate contribution is: Defines the L/C/R carrier as the first active state surface.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. The LCR Carrier.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines LCR as the primitive local address surface for the series.
- Proves the three-slot arity lower bound for center-plus-two-boundary data.
- Registers the binary shell profile 1,3,3,1 and the reversal action.
- Separates internal spinor-like local semantics from physical spin transport.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

Series Posture Inherited From FLCR-01

This paper inherits the FLCR-01 doctrine. Guardrails are automation controls: they govern AI agents, validators, build scripts, generated prose, and claim promotion workflows. They are not restrictions on human creativity or hypothesis formation.

The paper should therefore name the strongest intended reading of The LCR Carrier, display the parts already proved at this layer, and route the remaining distance as explicit open obligations. Those obligations are channels from the current prestate into later exploration.

The analog/workbook layer is also an access kit. It should preserve the local state in forms that can eventually be replayed without electricity, internet, computers, or paper. Later papers may work toward literal physics even where this local paper only supplies the finite or LCR-native carrier.

2. Source Routing

Primary routed sources:

- 01_LCR_Chain_Carrier.md
- supplements/01_INTERNAL_REASONING_SUPPLEMENT.md

Cross-corpus evidence classes:

- CAM_CLAIM_100_SCORE_AUDIT.md
- NSIT_TOOL_CLOSURE_MATRIX.md
- NSIT_REASONED_CLOSURE_PASS.md
- PAPER_REWORKS_CRYSTAL_PROJECTION.json
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **LCR state.** A triple (L, C, R) with left boundary, center, and right boundary coordinates.
- **Binary chart.** The finite carrier $\{0,1\}^3$ containing exactly eight states.
- **Shell grade.** The Hamming weight of the binary state, giving multiplicities 1,3,3,1.
- **Reversal.** The C2 action $\rho(L,C,R) = (R,C,L)$.

4. Formal Claims

Claim	Lane	Statement
Theorem 2.1	receipt_bound_internal_result	The LCR carrier is the minimal address-preserving carrier for one center and two distinguishable boundary slots.
Theorem 2.2	receipt_bound_internal_result	In the binary chart, reversal is an involution and partitions the carrier into fixed states and reversal pairs.
Proposition 2.3	standard_theorem_citation_bound_result	The shell profile of $\{0,1\}^3$ is the binomial profile 1,3,3,1.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 2.1	finite/executable result	The LCR carrier is the minimal address-preserving carrier for one center and two distinguishable boundary slots	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Theorem 2.2	finite/executable result	In the binary chart, reversal is an involution and partitions the carrier into fixed states and reversal pairs	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 2.3	formal construction	The shell profile of $\{0,1\}^3$ is the binomial profile 1,3,3,1	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-02 defines or uses The LCR Carrier as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Represent each local observation as (L,C,R), preserving left boundary, center, and right boundary.
2. Enumerate the binary chart $\{0,1\}^3$ and compute shell grade by Hamming weight.
3. Apply reversal $\rho(L,C,R)=(R,C,L)$ and record fixed points and two-state orbits.
4. Use the arity lower bound to show why fewer than three slots destroys either center identity or boundary distinction.

Proof Sketch

The finite proof is direct enumeration plus a minimality argument. There are exactly eight binary triples. Reversal applied twice returns the original triple, so it is a C2 action. Any two-slot representation has only two addresses and therefore must identify two of the three roles {left, center, right}; this violates the carrier contract.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is The LCR Carrier. Its safe consumption rule is:

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source object -> declared operation -> receipt/lane -> downstream import
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The unsafe consumption rule is:

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source phrase -> downstream identity claim
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That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

Recent External Quantum Evidence Intake

These rows add newly routed external physics papers as citation-bound or calibration-bound evidence. They are not CAM receipts unless a later tool run reconstructs their signatures.

Source	Lane	How It Helps This Paper	Boundary
EXT-RQM-CONSISTENT-2026: Quantum mechanics based on real numbers: A consistent description; DOI 10.1103/4k13-sdjh; arXiv 2503.17307	standard_theorem_citation_bound_result	Supports the carrier-layer distinction between an addressed physical state and the representational field chosen to describe it.	Do not infer that every LCR relabeling is physically equivalent; equivalence requires an explicit construction or adapter.

7. Evidence And Receipts

Evidence source	What it contributes
Paper 01 legacy body	Carrier enumeration, shell-2 doublet, local O8/double-cover route.
Internal supplement 01	Radius-1 CA grounding, arity lower bound, C2 action, binomial shell grading.
NSIT tool matrix	T5-T7 plus shell-2 doublet receipt and O8 spinor double-cover closure route.
CAM Claim 100 Score Audit	Paper 01 scored 94 as internally closed, with physical spinor transport deferred.

Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

Implementation source	Lane	Claim effect	Boundary
IMPL-LCR-TYPED-KERNEL	normal_form_result	Promotes L/C/R from notation to operational lane contract: LAdapter, CKernel, RChannel, strict policy gates, and lane classification.	This closes operational admission and policy routing; it does not by itself prove external physical identity.
IMPL-R30-LATTICE-THEOREM-REGISTRY	standard_theorem_citation_bound_result	Adds executable theorem registry rows for octonions, J3(O), chart-to-J3(O), exact n=3 SU(3) Weyl closure, Rule-30 8x8 transition, Niemeier routes, relational qubit closure, Monster/Pariah boundary, and computational ISA signatures. Verification: Targeted pytest passed: 11 Rule-30 normal-form/proof-shell tests, including address decomposition, centroid state, proof-shell dependency acyclicity, Niemeier obligation boundaries, scoped claim promotion, and full dependency closure.	Imports verifier-backed internal/formal results; transported physical interpretations remain lane-bound.

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.526710.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
carrier enumeration	normal_form_result	The LCR carrier action is closed as an addressable finite-state surface for its declared domain.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-1 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.510370 and mass sum=40.829567.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-02 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-02, FLCR-12, FLCR-22, FLCR-32; avg TarPit mass=0.511700; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
finite LCR carrier	receipt_bound_internal_result	The 2x2x2 carrier is closed as the finite local object consumed by later papers.	Physical interpretation of a carrier requires adapter rows.

Claim Posture After This Pass

FLCR-02 should now be read as a resolved-state contribution for The LCR Carrier at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR-11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.
- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal

normal-form intake.

- FLCR- 31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- The shell-2 doublet is an internal interface, not a measured particle claim.
- SU(2)-style language may be used as standard analogy only where explicitly lane-tagged.
- Later observer or physics papers must attach calibration or observer receipts before consuming this as physical spin.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 01/enumeration_action.

Original slot	Family	Lift	Role	Proof form
01	enumeration_action	0	order-1 slot-1: start the active one-dimensional action / enumerate the center state	enumeration proof and identity preservation

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 00 and reopened at original slot 01 (LCR Chain Carrier).

It must produce: formal result of original slot 01 plus the same-family lift path toward slot 11.

Semantic transition: select the active center and enumerate the addressable state space at the current lift.

Accepted formalism targets to bind in the journal rewrite:

- finite set enumeration
- ordered tuples and projections
- group actions on finite carriers
- minimality or lower-bound arguments

Slot	Family	Lift	Produced proof form
01	enumeration_action	0	enumeration proof and identity preservation

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to increase the state inventory: enumerate the local carrier, admission candidates, or applied reader objects before interpretation is added.

Primary Evidence Inputs

Source	Placement reason
01_LCR_Chain_Carrier.md	primary assigned source for the paper's formal spine
supplements/01_INTERNAL_REASONING_SUPPLEMENT.md	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
PAPER_REWORKS_CRYSTAL_PROJECTION.json	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NEW_PAPER_NEEDS.md	route relevant obligations into this paper's burden table: NP-04 F4 Encoder Universality And Exceptional Route Conditions
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.

- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

01	94	internally closed	LCR minimality, shell-2 doublet, local O8 double cover, Paper 03 carry-forward	physical spinor transport remains NP-10/physics bridge
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Paper-Specific NSIT Closure Rows

T5-T7 plus shell-2 doublet	batch_tool_unison_closure	f4_action.py, core.py	Papers 03 and 01	su3_closure_T5_T7_receipt.json 10/10; bijective_shell2_doublet_receipt.json 7/7	SU(3)/8x8 closure and single-tape doublet close.
O8 spinor SU(2) double cover	batch_tool_unison_closure	oloid_kinematic frame inversion semantics	Paper 01	o8_spinor_double_cover_closed_receipt.json 6/6	F^2 = -1 at 2pi; F^4 = +1 at 4pi.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Lane	Promote now	Bind next	Residue
Spinor/SU(2) Double-Cover Local Semantics	receipt_bound_internal_result	Promote local 2pi sign-flip / 4pi return semantics where the O8/frame-inversion receipt is attached.	Bind the O8 receipt and cite standard SU(2)->SO(3) double-cover semantics as standard analogy/support.	Physical observer, quantum measurement, and empirical spin claims remain routed to observer/physics calibration.

Binding Discipline

Each queue item is represented as a delta:

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local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
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This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Spinor Double Cover Local	This paper may claim local frame-inversion or spinor-style behavior only for its finite/internal carrier object.	O8 frame-inversion receipt plus standard SU(2)->SO(3) double-cover semantics.	Promote local 2pi sign-flip / 4pi return semantics as internal carrier semantics when the receipt is attached.	Do not promote to empirical spin, quantum measurement, or physical observer claims without external physics calibration.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-10: Paper 10 - T10 Master Receipt	81	Paper 10 proves the first stack-level receipt theorem in the CQECMPLX suite. Its object is not a new physical mechanism. Its object is the proof that Paper 00 and Papers 01-09 are bound into one inspectable and replay...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 24 — C-Form: KnightForge / N-Dimensional Chess Automata Gluon	77	C = the chess automata Gluon — the L-conjugate CA Gluon that generalizes knight moves to N-dimensional board operators. In the lattice_forge substrate, C is realized as the chess Gluon that: - The chess Gluon ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FINAL_FORMAL_PAPER: Complete Formal Claims: The Folded Strand	75	We present the complete closed-form claim set of the CQE_CMLPX corpus: - 33 papers = 33 folding operations on a self-complementary strand - 144 tools = cumulative analog kit = digital twin surface - 135 digital twins ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
SUMMARY-V-32-Theorems-Registry: Summary Paper V — The 32 Theorems in One Table: Closed-Form Registry	75	This paper is the complete closed-form registry of all 32 theorems in the CQE_CMLPX corpus. Each theorem is stated precisely, given its formal context (where it is proven), and listed with its verifier. The table ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-10.25: Paper 10.25 - Toolkit for the T10 Master Receipt	73	Paper 10.25 describes the tools for reviewing the T10 master receipt. These tools expose receipt binding, transport classification, local witness replay, and lookup cache materialization. They do not close the open li...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	34 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR-02
Legacy source slot(s)	01
Ribbon role	order-1 slot-1: start the active one-dimensional action / enumerate the center state
Proof form	enumeration proof and identity preservation
Received state	state emitted by prior slot 00 and reopened at original slot 01 (LCR Chain Carrier)
Produced state	formal result of original slot 01 plus the same-family lift path toward slot 11

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 2.1	receipt_bound_internal_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	The LCR carrier is the minimal address-preserving carrier for one center and two distinguishable boundary slots.

Claim	Lane	Evidence to attach	Boundary
Theorem 2.2	receipt_bound_internal_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	In the binary chart, reversal is an involution and partitions the carrier into fixed states and reversal pairs.
Proposition 2.3	standard_theorem_citation_bound_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	The shell profile of $\{0,1\}^3$ is the binomial profile 1,3,3,1.

Figure Plan

Figure	Purpose	Caption
FLCR-02-F1	State enumeration chart	Schematic showing how FLCR-02 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-02-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-02-F3	Same-family lift context	Diagram placing this paper in the original 00 - 79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-02-F1: State enumeration chart

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received state
-> State enumeration chart
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
-> remaining residue

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Table Plan

Table	Purpose
FLCR-02-T1	State inventory table
FLCR-02-T2	Claim-lane/evidence-boundary table
FLCR-02-T3	Archive evidence card and source-backed upgrade table
FLCR-02-T4	Workbook replay and falsifier table

Worked Example

Enumerate the local carrier or admission candidates and classify each row before interpretation. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.
addressable slot	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.
center state	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.
enumeration	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.
finite domain	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the enumeration_action proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-02-A: Evidence Package

The appendix records:

- 1. Legacy source excerpts or source-card references used in the final claim ledger.
- 2. Receipt and verifier paths, including pass/fail/partial status.
- 3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
- 4. Standards-window and NSIT evidence used for conformance or routing.
- 5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-02 (The LCR Carrier) occupies the enumeration_action role at lift depth 0. The paper receives state emitted by prior slot 00 and reopened at original slot 01 (LCR Chain Carrier). It produces formal result of original slot 01 plus the same-family lift path toward slot 11. The state transition is: select the active center and enumerate the addressable state space at the current lift.

The paper-local proof form is:

enumeration proof and identity preservation

The integration result is an identity-preserving inventory of admissible state objects. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
finite set enumeration	Formal carrier for the paper-local result.
ordered tuples and projections	Formal carrier for the paper-local result.
group actions on finite carriers	Formal carrier for the paper-local result.
minimality or lower-bound arguments	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 2.1	receipt_bound_internal_result	The LCR carrier is the minimal address-preserving carrier for one center and two distinguishable boundary slots.
Theorem 2.2	receipt_bound_internal_result	In the binary chart, reversal is an involution and partitions the carrier into fixed states and reversal pairs.
Proposition 2.3	standard_theorem_citation_bound_result	The shell profile of $\{0,1\}^3$ is the binomial profile 1,3,3,1.

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	01_LCR_Chain_Carrier.md, supplements/01_INTERNAL_REASONING_SUPPLEMENT.md	routed evidence
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-02-F1: State enumeration chart	Visualizes the proof object, routing, or boundary.
FLCR-02-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-02-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-02-T1: State inventory table	Records claim lanes, evidence, sources, or falsifiers.

Table	Role
FLCR-02-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-02-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-02-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.